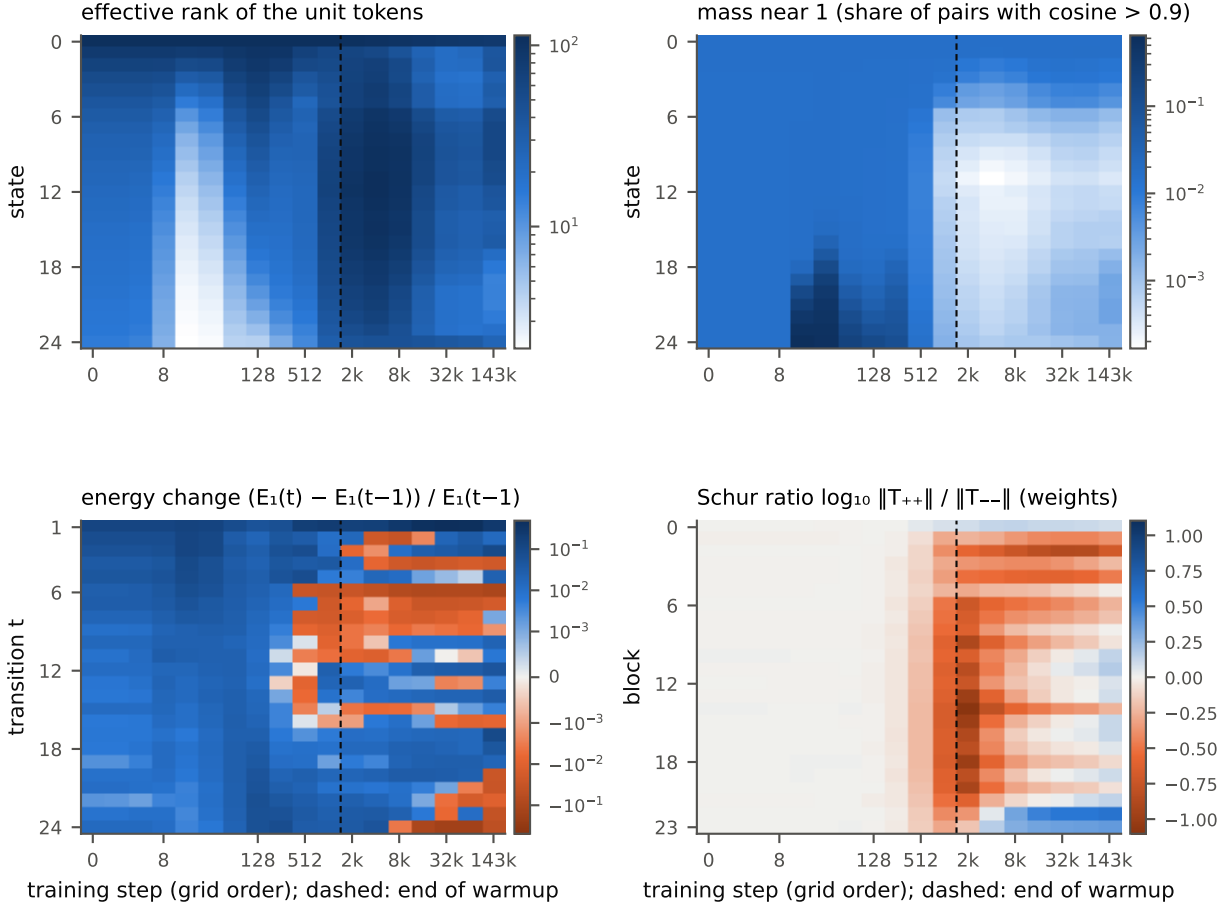
effective rank of the unit tokens
state
0
6
12
18
24
0
8
128
512
2k
8k
32k
143k
10^2
10^1
mass near 1 (share of pairs with cosine > 0.9)
state
10^{-1}
10^{-2}
10^{-3}
energy change (E₁(t) − E₁(t−1)) / E₁(t−1)
transition t
1
6
12
18
24
10^{-1}
10^{-2}
10^{-3}
0
-10^{-3}
-10^{-2}
-10^{-1}
training step (grid order); dashed: end of warmup
Schur ratio log₁₀ ‖T₊₊‖ / ‖T₋₋‖ (weights)
block
0
6
12
18
23
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
training step (grid order); dashed: end of warmup